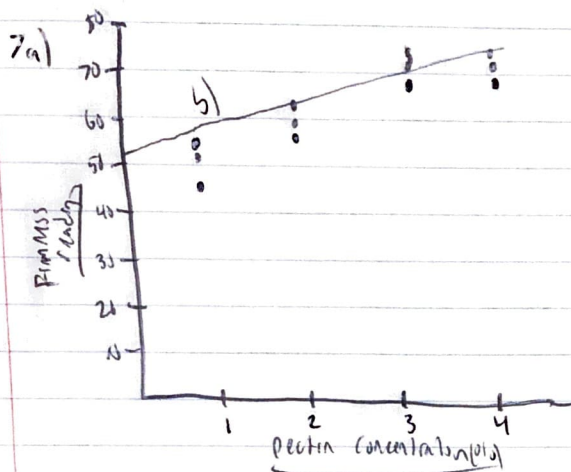


Alex Caramanna

4-13-22

Stat 434 HW10

Ch 11: 7, 8, 11, 12, 13, 14, 19, 30, 31, 46a, 46b



x	0	1
y	46.4, 50.2, 51.3	56.48, 59.34, 62.97

x	2	3
y	67.41, 70.78, 73.67	68.13, 72.95, 72.54

b) obtain least-squares estimates for the parameters, plot them on the scatter diagram

$$\beta_1 = \frac{S_{xy}}{S_{xx}} \quad \beta_0 = \bar{y} - \beta_1 \bar{x} \quad \bar{x} = 1.5: \bar{y} = 62.576$$

$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = 111.32$$

$$S_{xx} = \sum (x_i - \bar{x})^2 = 15$$

$$\frac{111.32}{15} = 7.42 = \beta_1$$

$$51.44 + 7.42x$$

$$\beta_0 = 62.576 - 7.42(1.5)$$

$$= 62.576 - 11.132 = 51.44$$

c) firmness does appear to be in a constant (linear) relationship w/ pectin concentration, at least according to the LSEs. The raw data appears to level off slightly at concentrations of 200 and 300.

d) $\hat{y} = 51.44 + 7.42x$ where x = pectin concentration

if $x = 1.5$, $\hat{y} = 62.57$

1. use 4.9 regression line

$$\hat{y} = 6.03 - 0.0170x \quad \text{where } y = \text{lifetime, } x = \text{drill speed}$$

a)	x	60	80	100	120	140
y		5.01	4.67	4.33	3.99	3.65

b)	blower or smaller than actual	bigger	smaller	maxi smaller	not accurate	not accurate
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c) perhaps taking the log of the drill speed or square roots would allow for a better (linear) fit.

* linear relationship occurs at drill speeds 2120

$n=98$

$$s_{\beta_1}^2 = \frac{\sum (y_i - \hat{y}_i)^2}{n-2} = \frac{11.73}{96} = 0.1211$$

Refer to #7.

a) 95% CI for β_1 : $\hat{\beta}_1 \pm t_{\alpha/2, n-2} \cdot s_{\beta_1} \rightarrow 7.42 \pm 1.984 \cdot 0.348$

$7.42 \pm 1.984 (0.1153) = (7.191, 7.649)$

b) interpretation of $H_0: \beta_1 = 0$

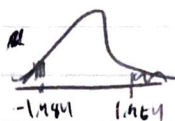
If $\beta_1 = 0$, then it would suggest there is no relationship between pocket concentration and the firmness of the current pilotages

c) Test hypothesis: $H_0: \beta_1 = 0$ vs $H_a: \beta_1 \neq 0$

$T = \frac{\hat{\beta}_1 - 0}{s_{\beta_1}}$

CV: $t_{n-2, \alpha/2} = t_{96, 0.025} = 1.984$

$LS = \frac{7.42}{0.1153} = 64.35$



Reject null hypothesis, conclude $\beta_1 \neq 0$ at $\alpha = 0.05$

d) p-value: $P(t_1 \geq 64.35)$ is much less than $\alpha = 0.05$, so there is sufficient evidence to reject the null hypothesis

Bal 95% confidence CI for β_0 : $\hat{\beta}_0 \pm t_{\alpha/2, n-2} \cdot s_{\beta_0}$

$51.44 \pm 1.984 (3.361) = (14.761, 58.119)$

$s_{\beta_0}^2 = \frac{s_e^2}{n} \left(\frac{1}{n} + \frac{\bar{y}^2}{\sum x_i^2} \right)$